Grafos Hamiltonianos e Grafos Eulerianos

Zenilton Patrocínio

Grafo Hamiltoniano

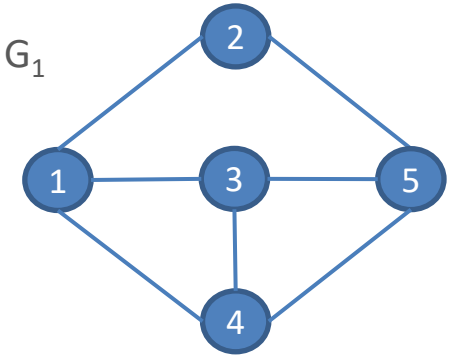
Um **caminho hamiltoniano** é um caminho que passa por cada vértice de um grafo exatamente uma vez.

Um **ciclo hamiltoniano** é um caminho hamiltoniano que retorna ao vértice inicial (isto é, um caminho fechado).

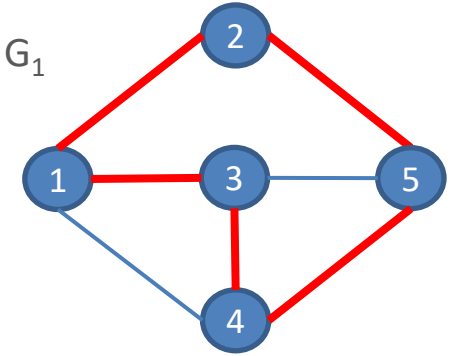
Um **grafo** é dito **hamiltoniano** se possuir um **ciclo hamiltoniano**.

Um **grafo** é dito **semi-hamiltoniano** se possuir um **caminho hamiltoniano**. Logo, um grafo hamiltoniano é também semi-hamiltoniano.

Grafo Hamiltoniano – Exemplo



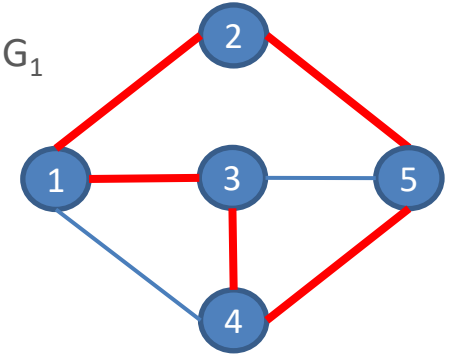
Grafo Hamiltoniano – Exemplo



Hamiltoniano

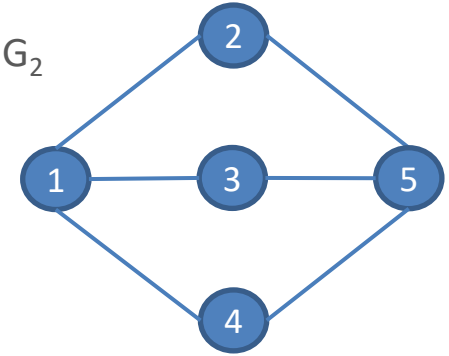
Grafo Hamiltoniano – Exemplo

G_1



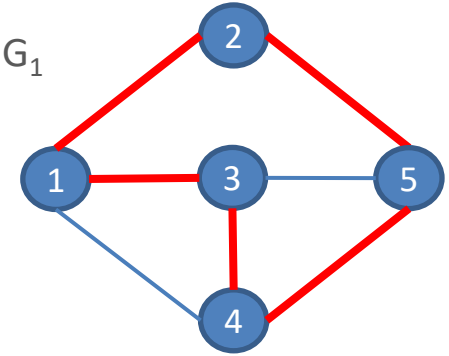
Hamiltoniano

G_2



Grafo Hamiltoniano – Exemplo

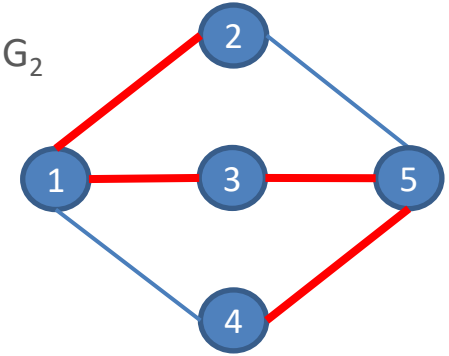
G_1



Hamiltoniano

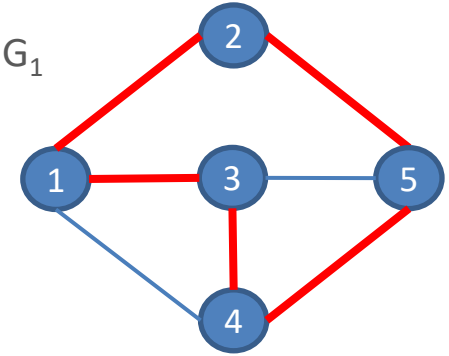
Semi-hamiltoniano

G_2



Grafo Hamiltoniano – Exemplo

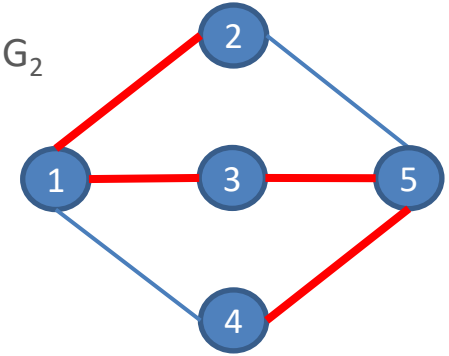
G_1



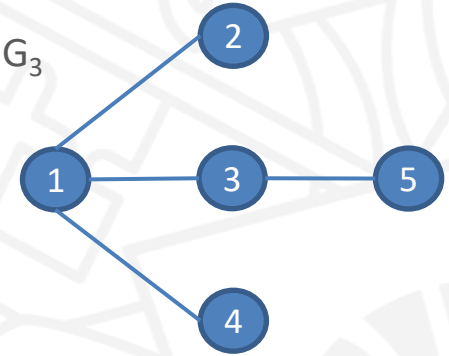
Hamiltoniano

Semi-hamiltoniano

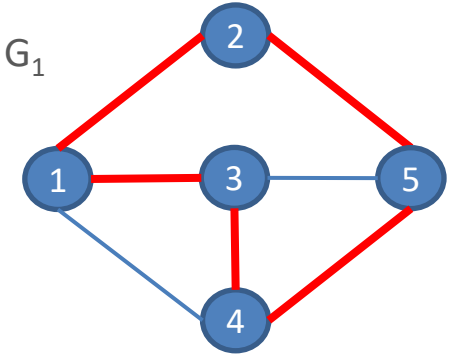
G_2



G_3

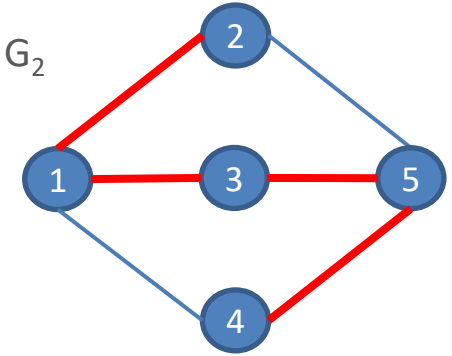


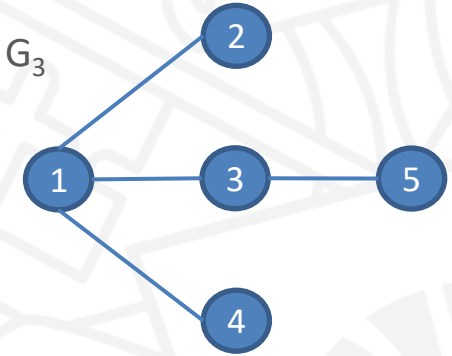
Grafo Hamiltoniano – Exemplo



Hamiltoniano

Semi-hamiltoniano





Não-hamiltoniano

Grafo Hamiltoniano – Condição Suficiente

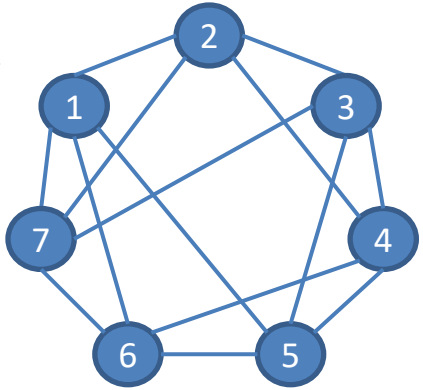
Um grafo simples G com n (≥ 3) vértices é hamiltoniano, se o grau de cada um de seus vértices $d(v) \geq n/2$, $\forall v \in V(G)$. (Teorema de Dirac)

Um grafo simples G com n (≥ 3) vértices é hamiltoniano, se, para cada par de vértices não adjacentes v e w, a soma de seus graus $d(v) + d(w) \geq n$, $\forall \{v,w\} \notin E(G)$. (Teorema de Ore)

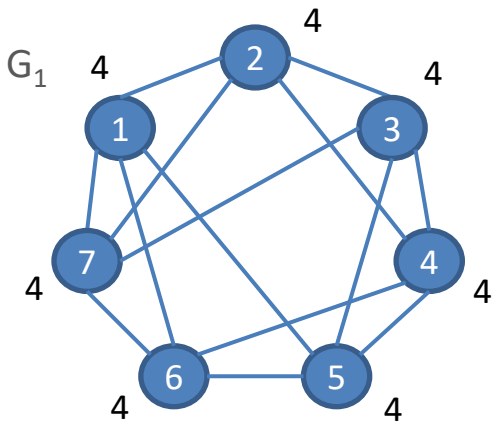
Se o fecho hamiltoniano de G for um grafo completo, então G é hamiltoniano. Fecho hamiltoniano de uma grafo é obtido adicionando-se arestas, enquanto for possível, entre vértices não adjacentes cuja soma de graus $\geq n$. (Teorema de Bondy & Chvátal)

Grafo Hamiltoniano – Condição Suficiente

G_1



Grafo Hamiltoniano – Condição Suficiente



Atende ao Teorema de Dirac

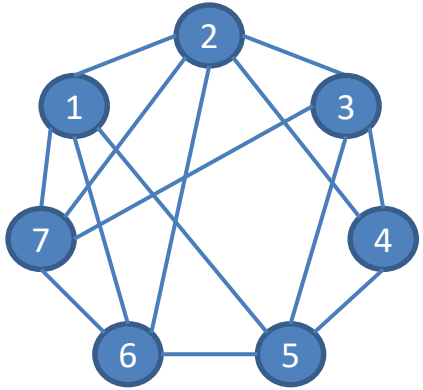
Atende ao Teorema de Ore

Atende ao Teorema de Bondy & Chvátal

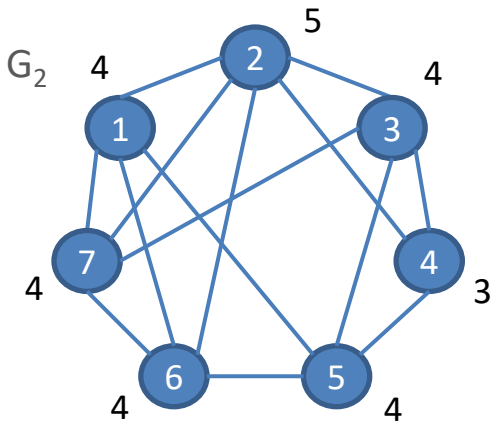
G_1 é hamiltoniano

Grafo Hamiltoniano – Condição Suficiente

G_2



Grafo Hamiltoniano – Condição Suficiente



Não atende ao Teorema de Dirac

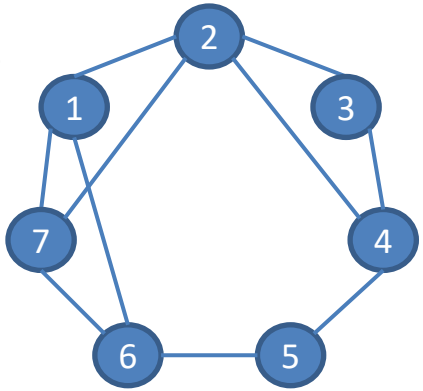
Atende ao Teorema de Ore

Atende ao Teorema de Bondy & Chvátal

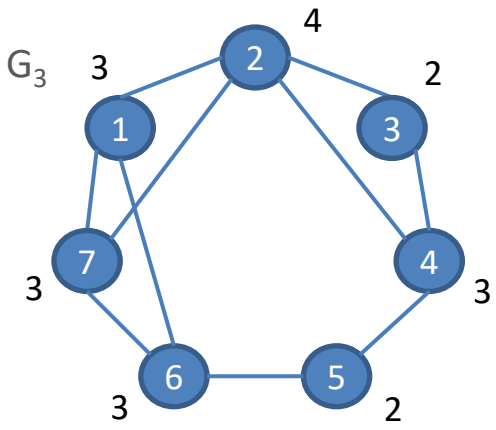
G_2 é hamiltoniano

Grafo Hamiltoniano – Condição Suficiente

G_3



Grafo Hamiltoniano – Condição Suficiente



Não atende ao Teorema de Dirac

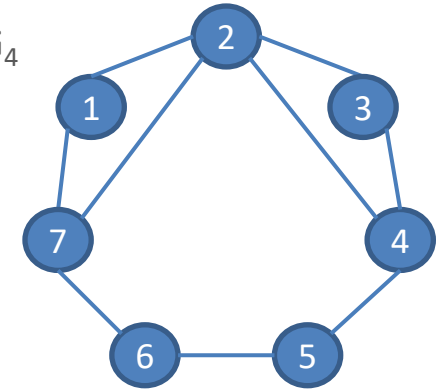
Não atende ao Teorema de Ore

Atende ao Teorema de Bondy & Chvátal

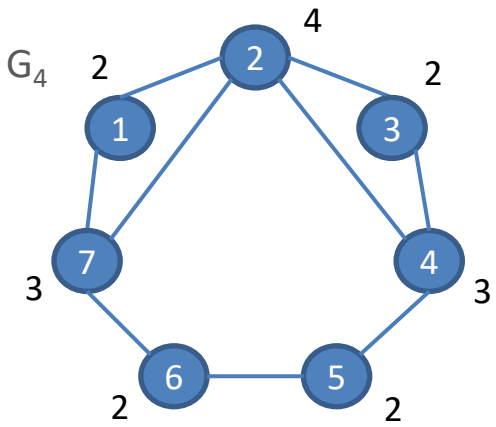
G_3 é hamiltoniano

Grafo Hamiltoniano – Condição Suficiente

G_4



Grafo Hamiltoniano – Condição Suficiente



Não atende ao Teorema de Dirac

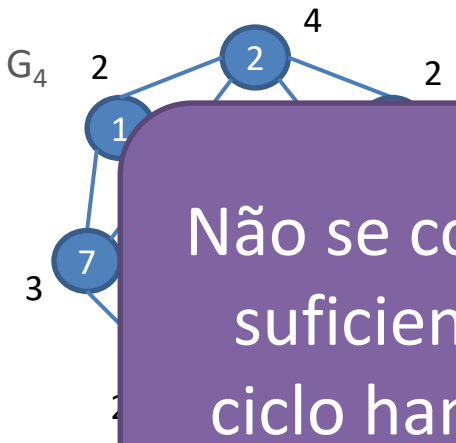
Não atende ao Teorema de Ore

Não atende ao Teorema de Bondy & Chvátal

PORÉM

G_4 é hamiltoniano

Grafo Hamiltoniano – Condição Suficiente



Não se conhece uma condição necessária e suficiente trivial para a existência de um ciclo hamiltoniano em um grafo qualquer.

Grafo Euleriano

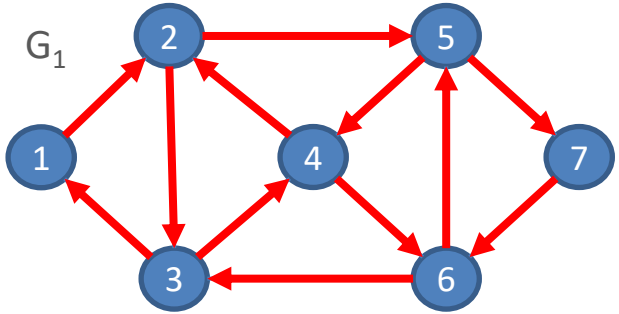
Um trajeto euleriano é um trajeto que passa por cada aresta de um grafo exatamente uma vez.

Um ciclo euleriano é um trajeto euleriano que começa e termina no mesmo vértice (isto é, um trajeto fechado).

Um grafo é dito euleriano se possuir um **ciclo euleriano**.

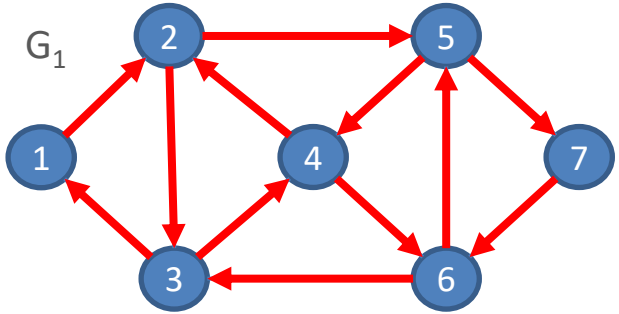
Um grafo é dito semi-euleriano se possuir um **trajeto euleriano**. Logo, um grafo euleriano é também semi-euleriano.

Grafo Euleriano – Exemplo



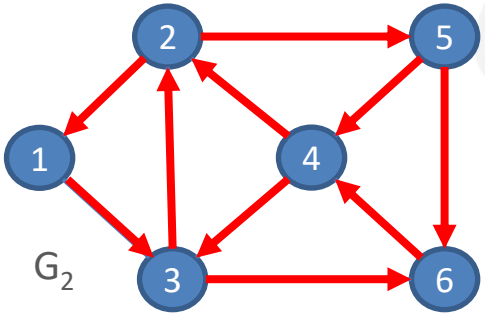
Euleriano

Grafo Euleriano – Exemplo

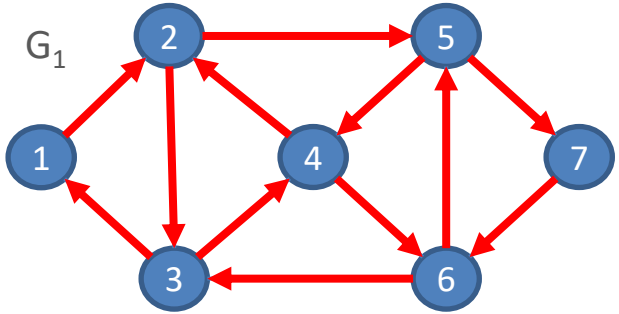


Euleriano

Semi-euleriano

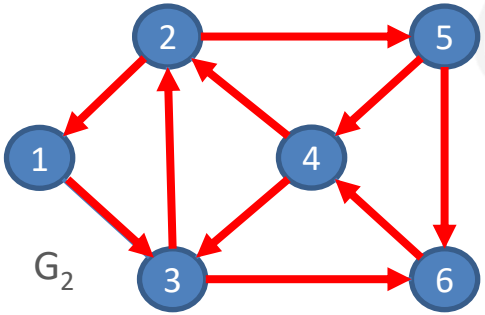


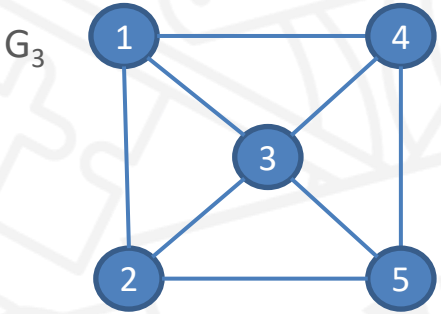
Grafo Euleriano – Exemplo



Euleriano

Semi-euleriano





Não-euleriano

Grafo Euleriano – Condição Suficiente

Um grafo conexo é euleriano se e somente se todos os seus vértices possuírem grau par. (Teorema de Euler)

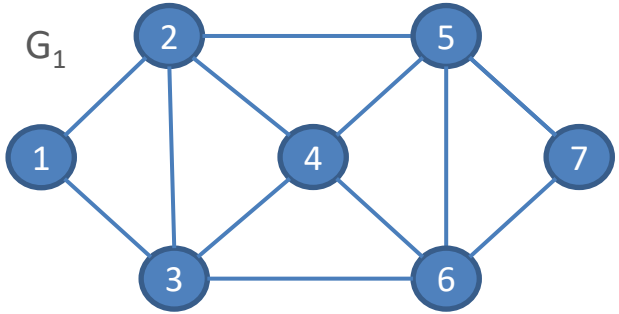
Um grafo conexo é não-euleriano se existirem dois os mais vértices de grau ímpar.

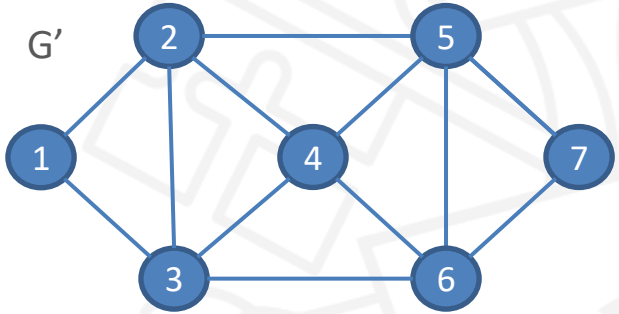
Um grafo conexo é semi-euleriano se e somente se existem exatamente dois vértices de grau ímpar.

Método de Fleury – Algoritmo

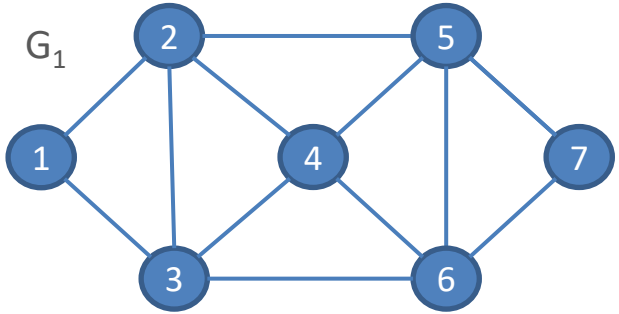
1. <u>se</u> V(G) possuir 3 ou mais vértices de grau ímpar <u>então</u> **PARE**;
2. Seja G’ = (V’, E’) tal que V’ $\leftarrow$ V(G)e E’ $\leftarrow$ E(G); // Inicializar grafo auxiliar
3. Selecionar vértice inicial $v \in$ V’ (escolher v cujo grau seja ímpar, se houver)
4. <u>enquanto</u> E’ $\neq \varnothing$ <u>efetuar</u>
   a. se $d(v) > 1$ então
      Selecionar aresta $\{v, w\}$ que não seja ponte em G’
   b. senão
      Selecionar a única aresta $\{v, w\}$ disponível em G’
   c. $v \leftarrow w$; E’ $\leftarrow$ E’ – $\{v, w\}$; // Caminhar de v para w e eliminar aresta

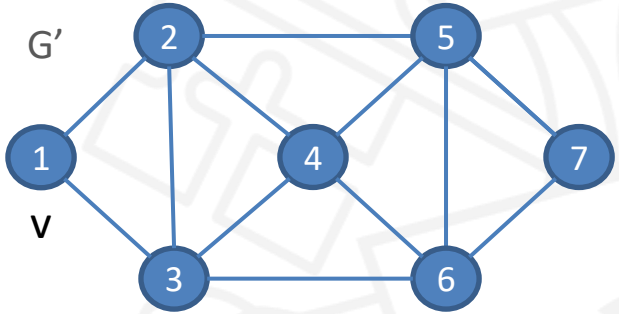
Método de Fleury – Exemplo 1





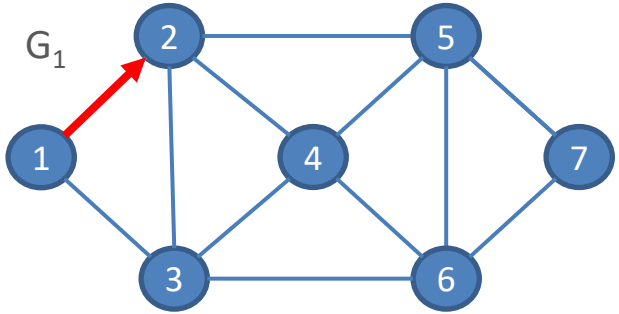
Método de Fleury – Exemplo 1

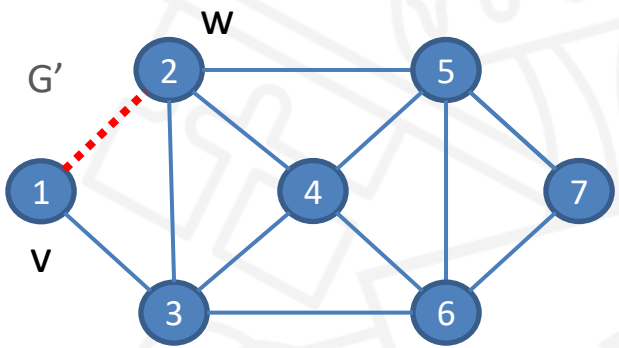




1

Método de Fleury – Exemplo 1





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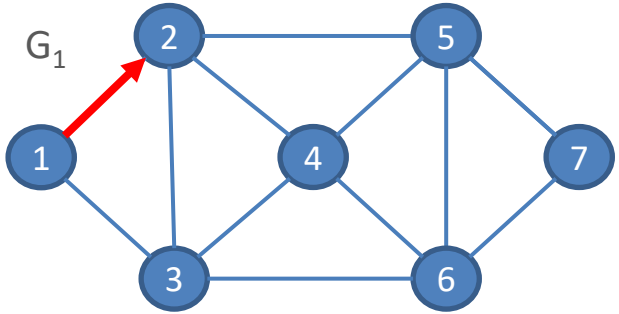
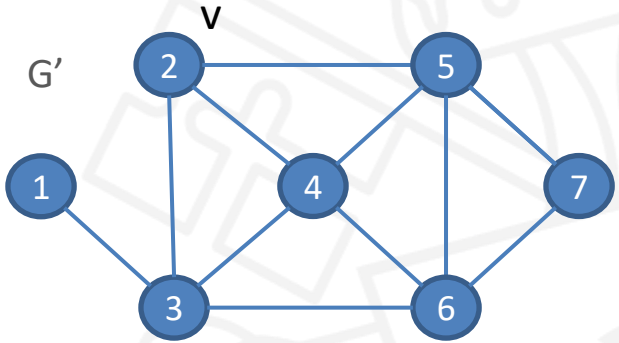
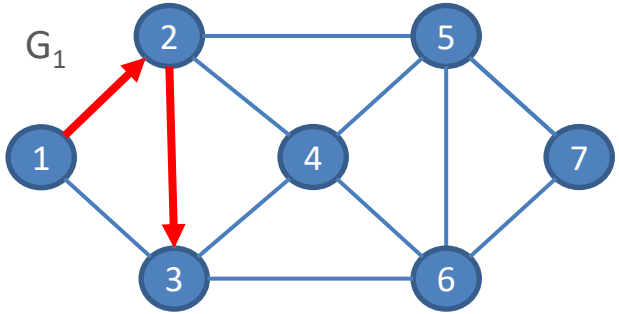
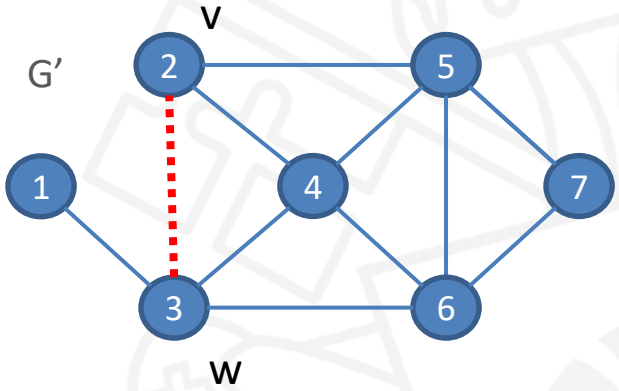




1 / 2

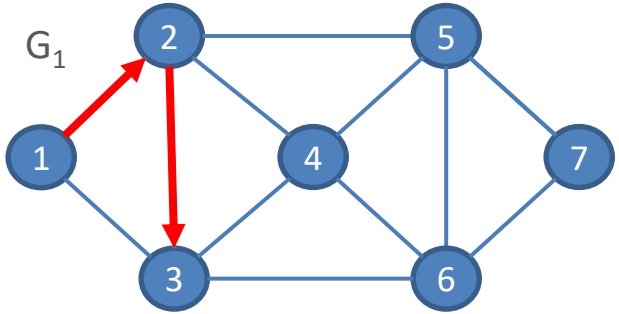
Método de Fleury – Exemplo 1

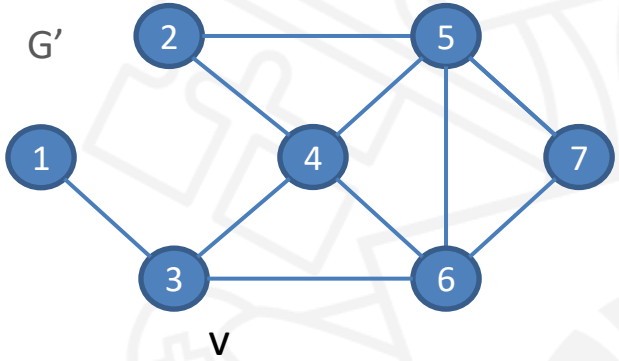




1 / 2

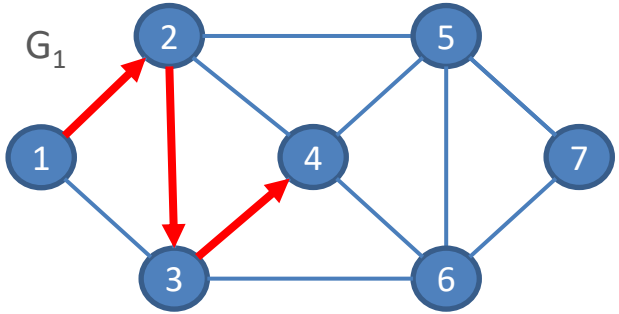
Método de Fleury – Exemplo 1

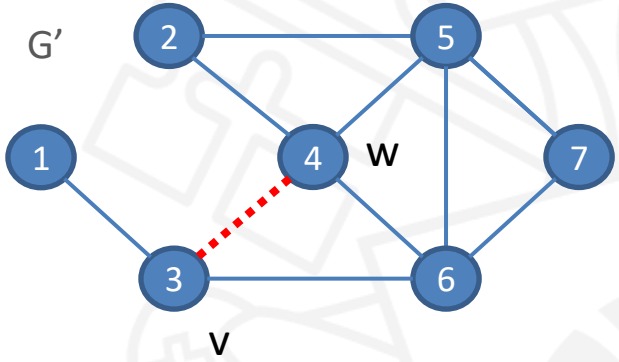




1 / 2 / 3

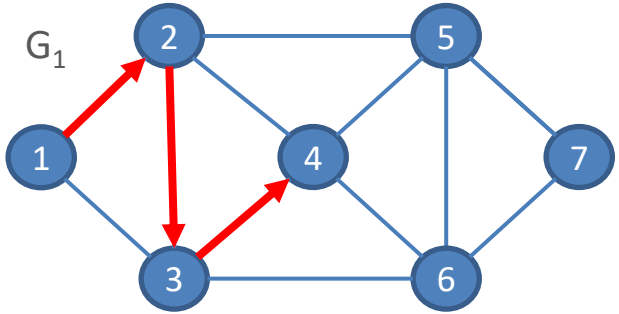
Método de Fleury – Exemplo 1

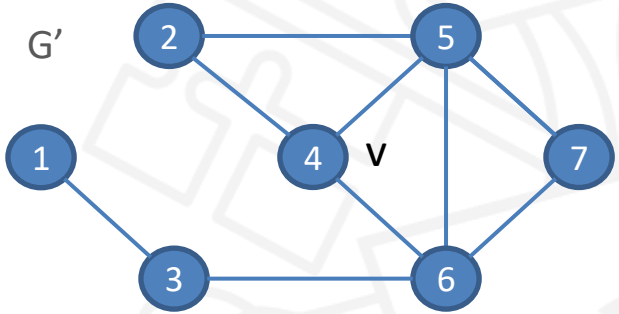




1 / 2 / 3

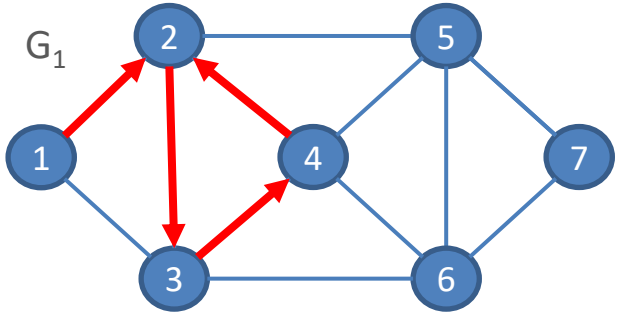
Método de Fleury – Exemplo 1

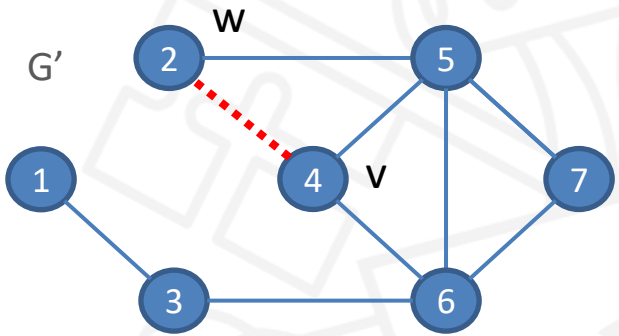




1 / 2 / 3 / 4

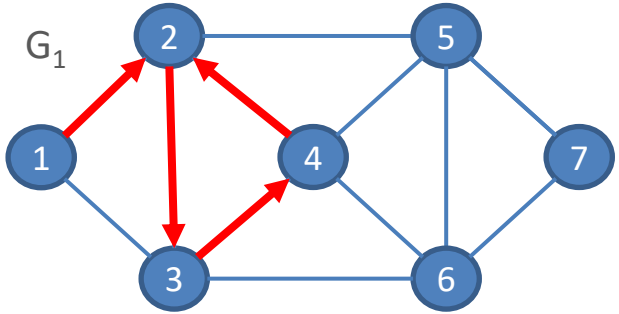
Método de Fleury – Exemplo 1

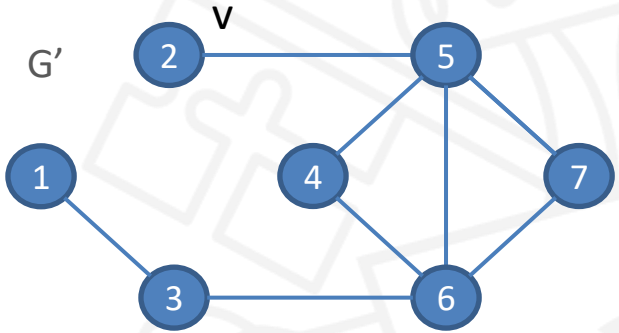




1 / 2 / 3 / 4

Método de Fleury – Exemplo 1





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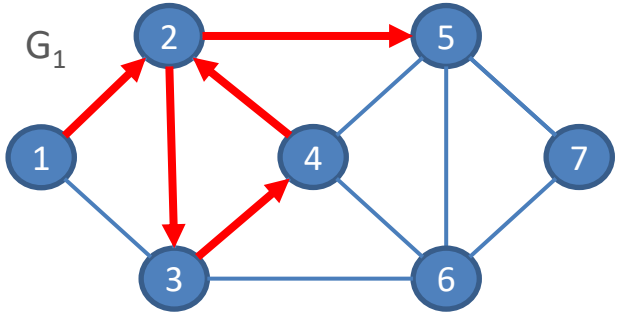
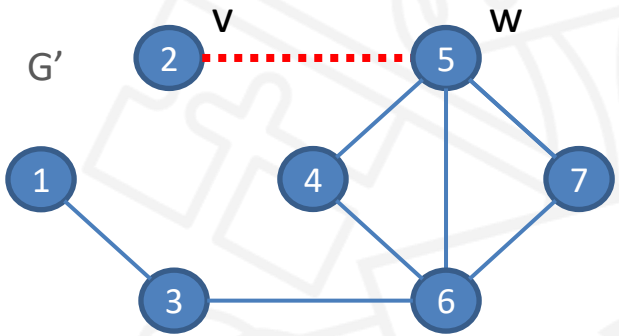
Método de Fleury – Exemplo 1





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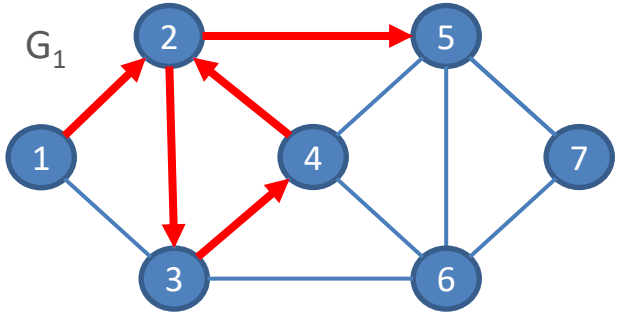
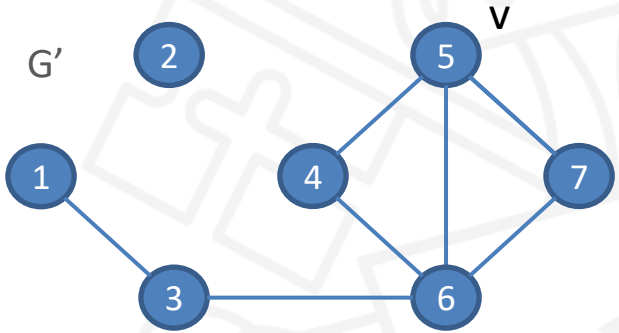
Método de Fleury – Exemplo 1





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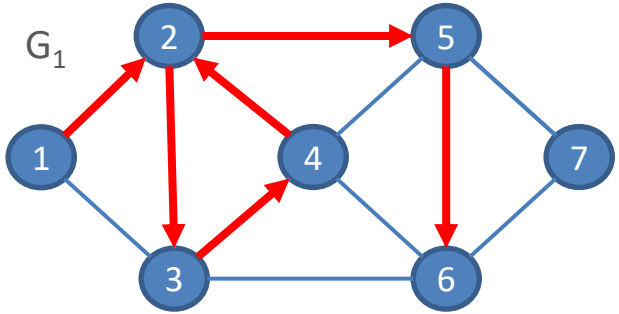
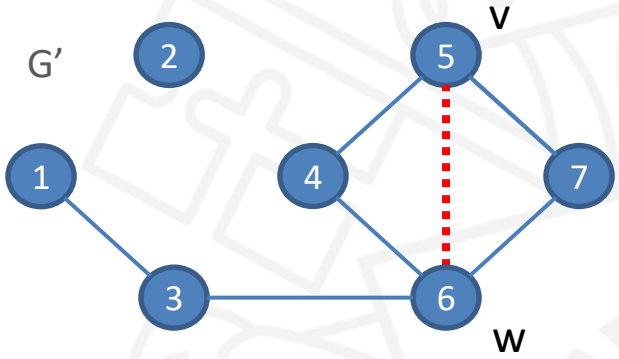
Método de Fleury – Exemplo 1





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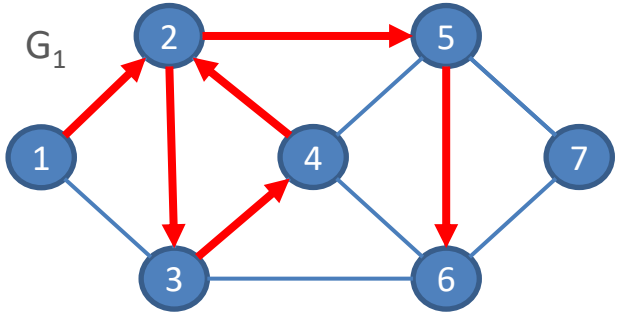
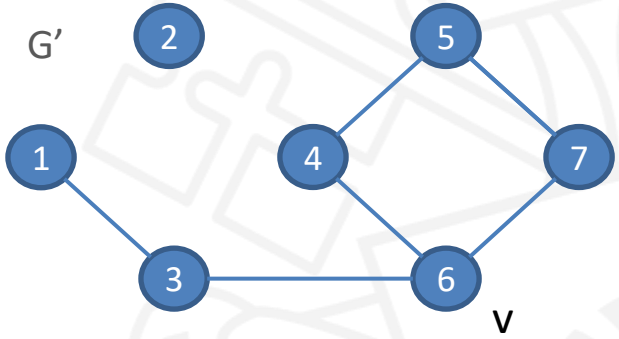
Método de Fleury – Exemplo 1





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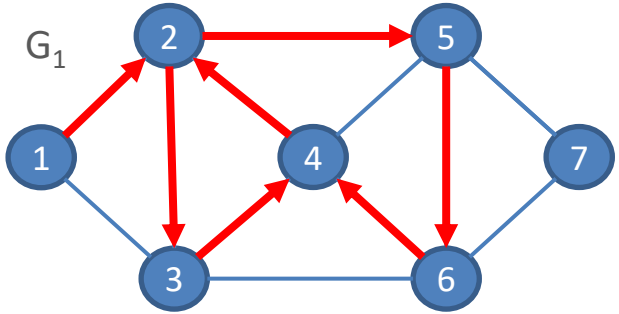
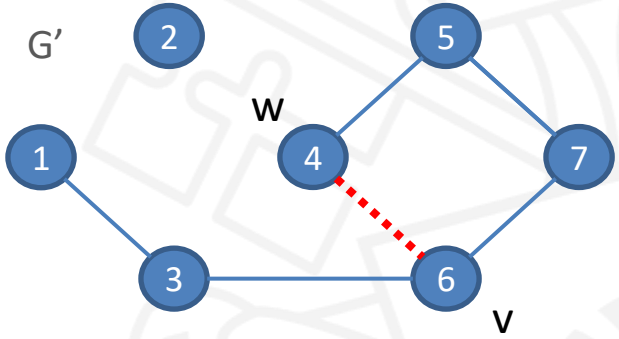
Método de Fleury – Exemplo 1





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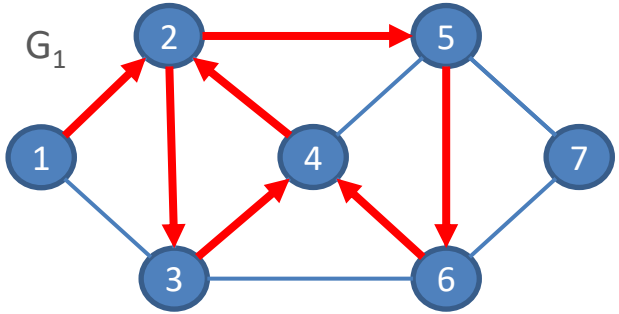
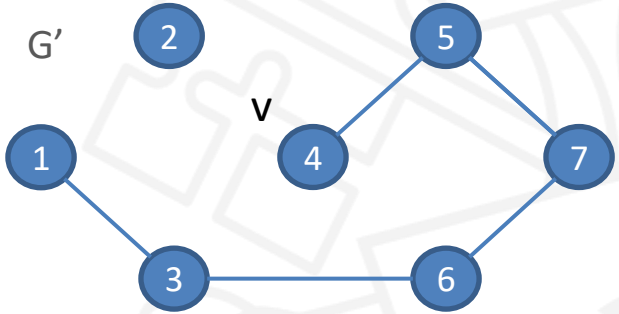
Método de Fleury – Exemplo 1





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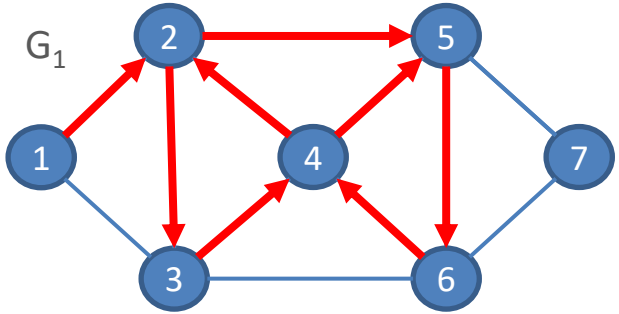
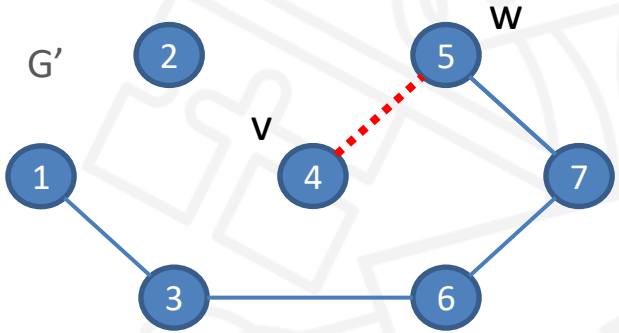
Método de Fleury – Exemplo 1





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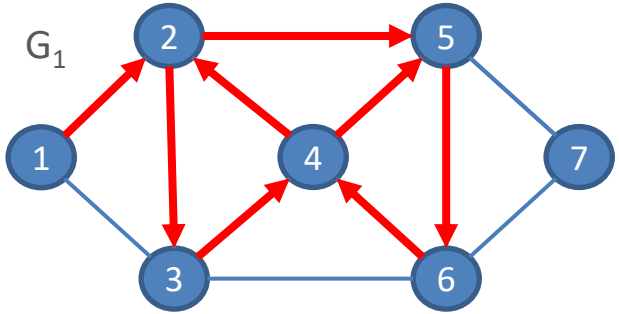
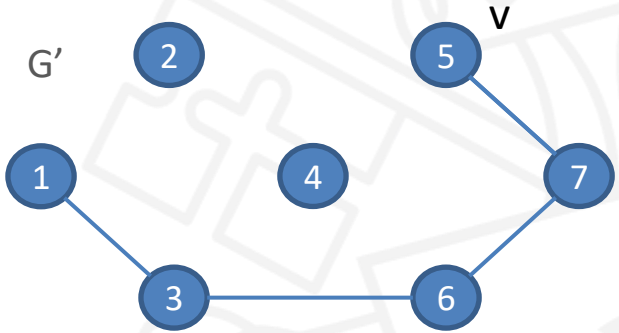
Método de Fleury – Exemplo 1





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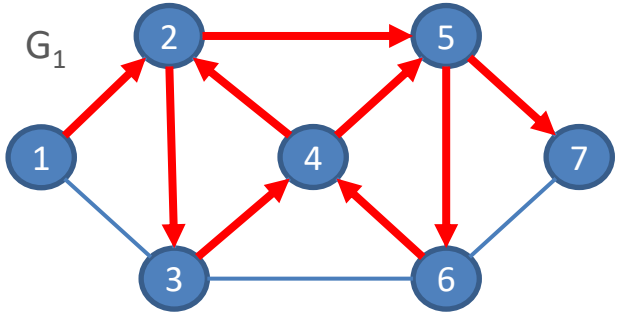
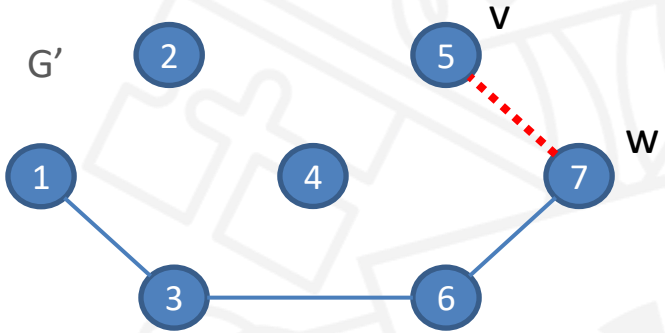
Método de Fleury – Exemplo 1





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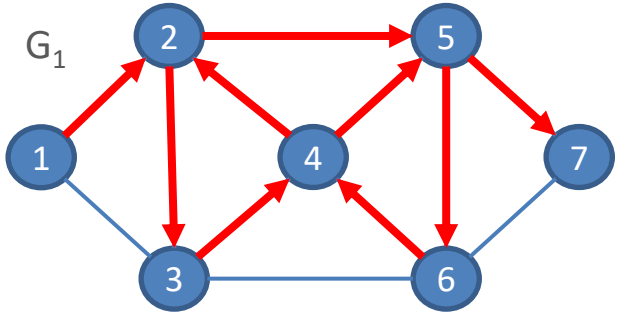
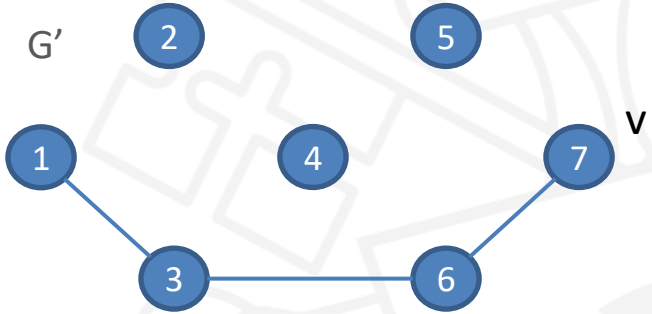
Método de Fleury – Exemplo 1





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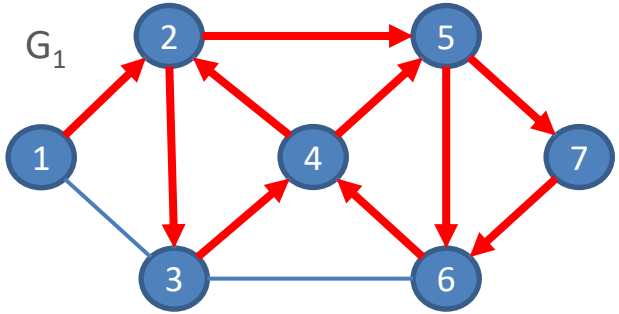
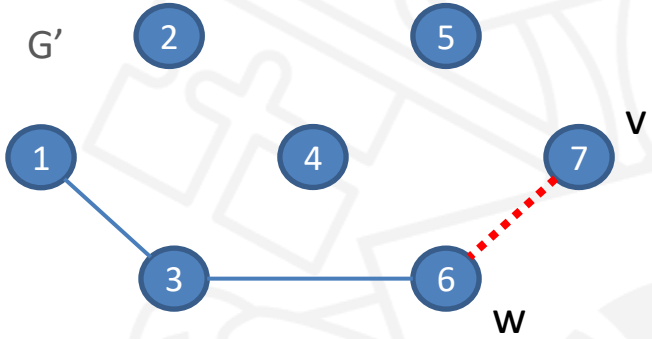
Método de Fleury – Exemplo 1





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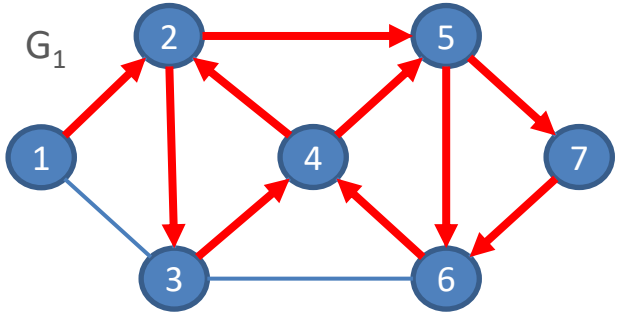
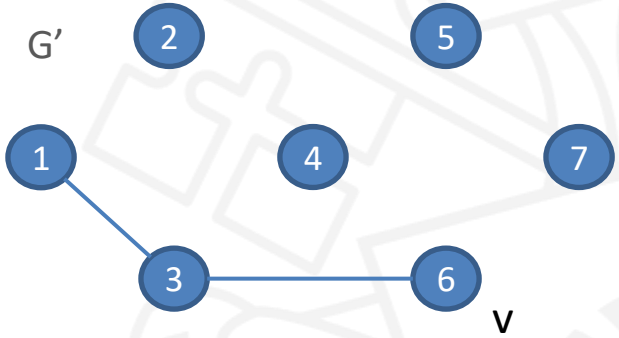
Método de Fleury – Exemplo 1





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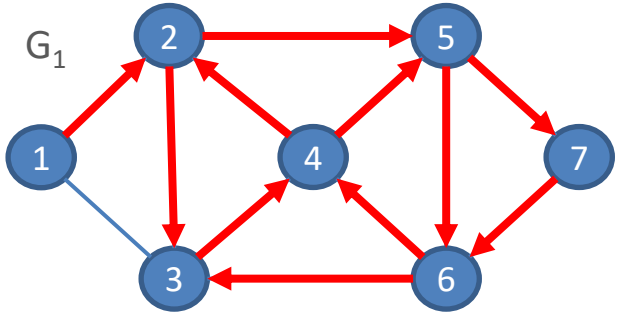
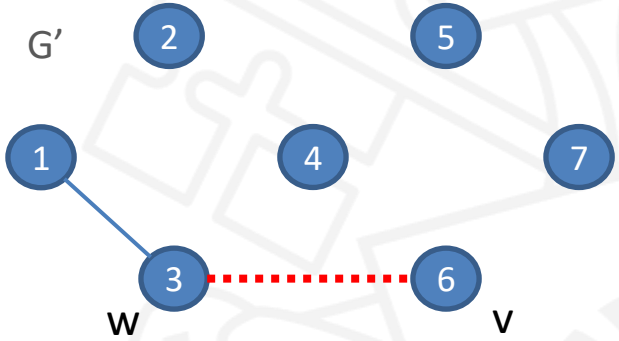
Método de Fleury – Exemplo 1





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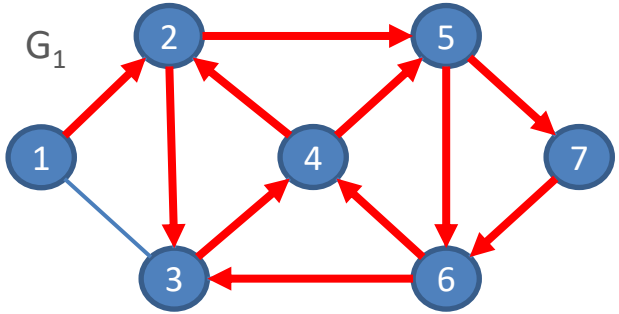
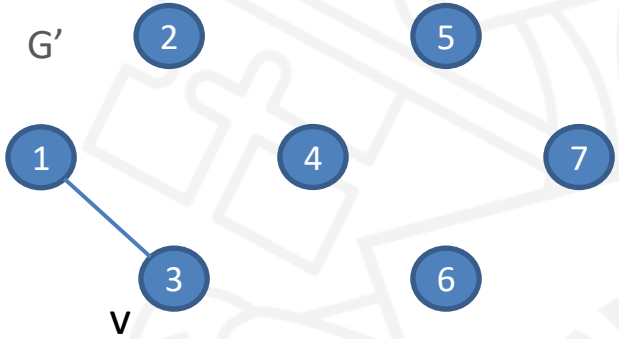
Método de Fleury – Exemplo 1





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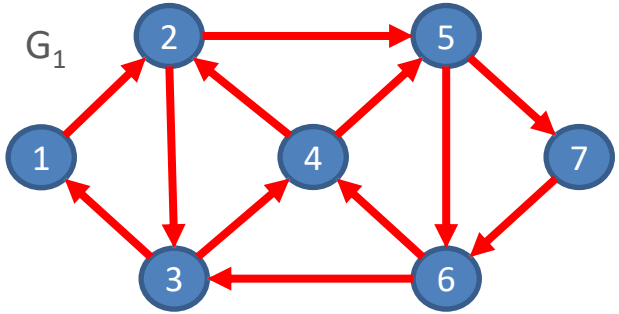
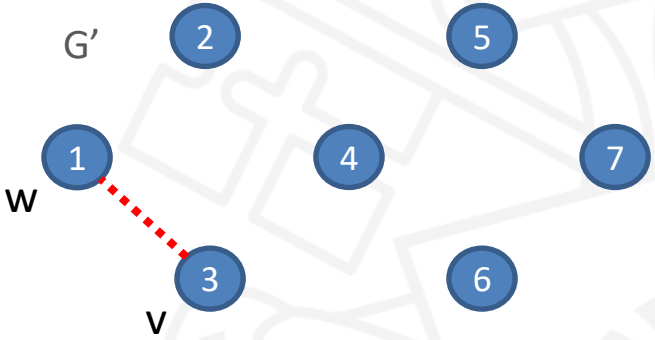
Método de Fleury – Exemplo 1





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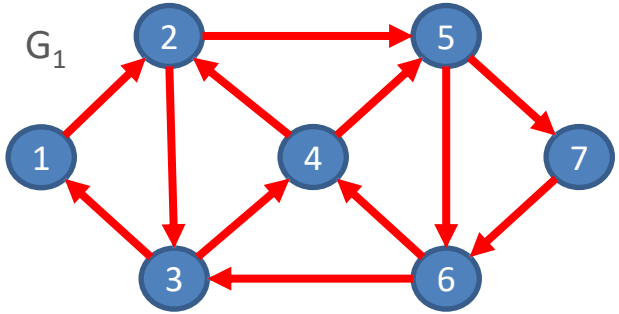
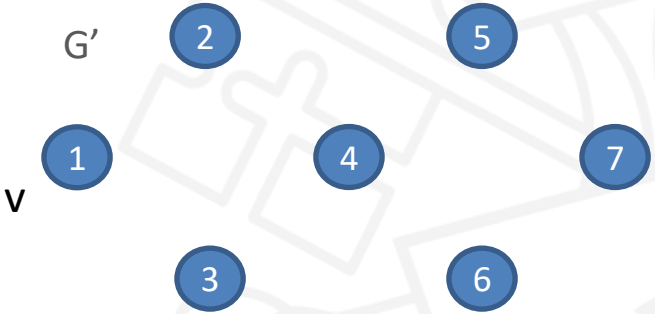
Método de Fleury – Exemplo 1





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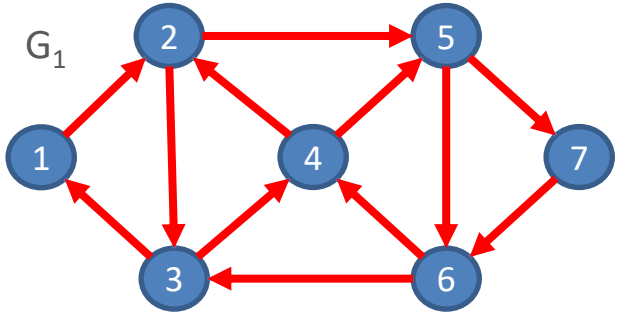
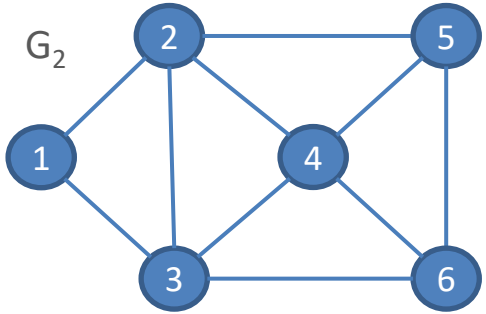
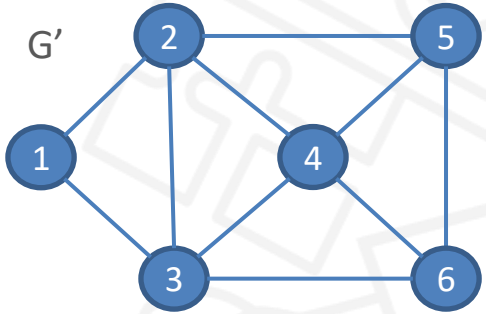
Método de Fleury – Exemplo 1



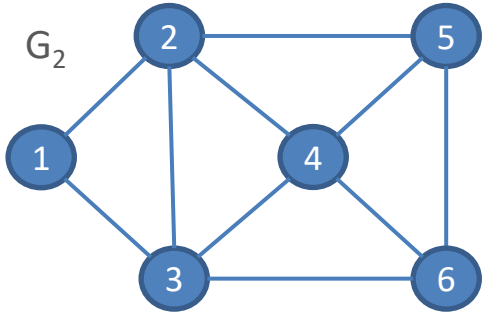
1 / 2 / 3 / 4 / 2 / 5 / 6 / 4 / 5 / 7 / 6 / 3 / 1 → **Ciclo euleriano**

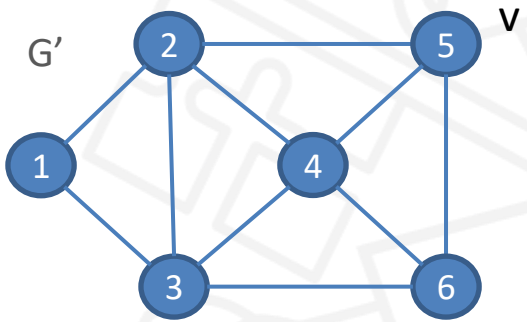
Método de Fleury – Exemplo 2





Método de Fleury – Exemplo 2





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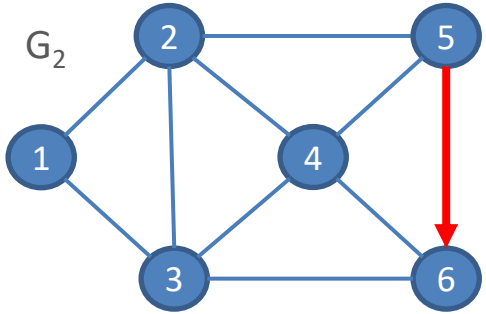
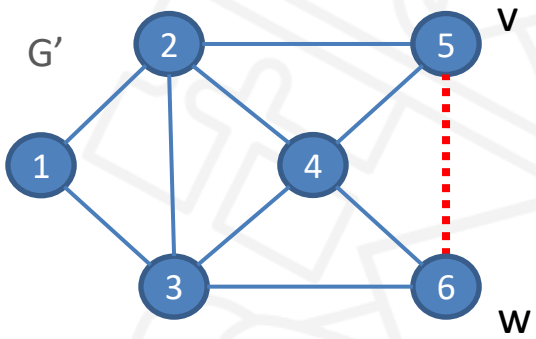
Método de Fleury – Exemplo 2





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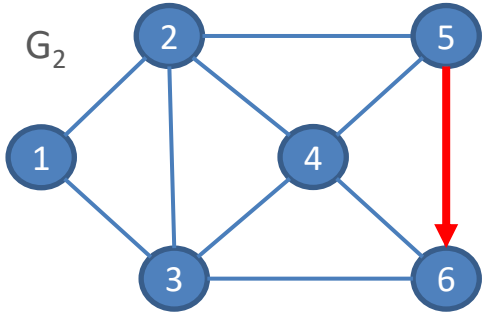
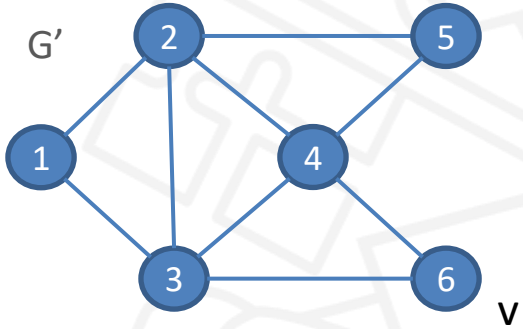
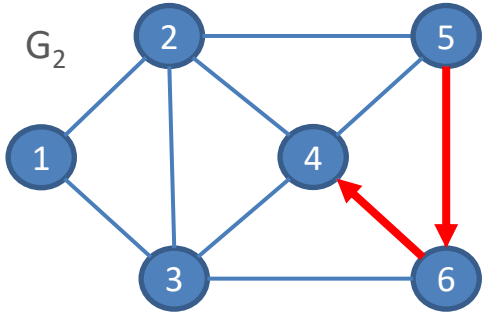
Método de Fleury – Exemplo 2

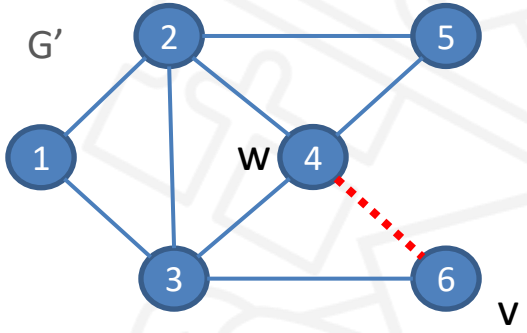




5 / 6

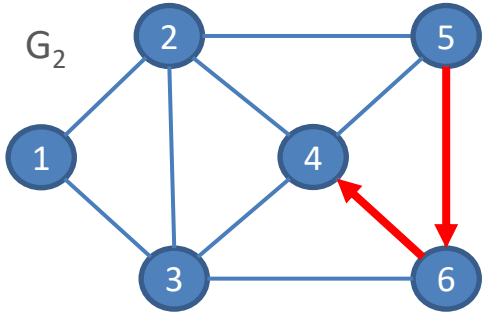
Método de Fleury – Exemplo 2

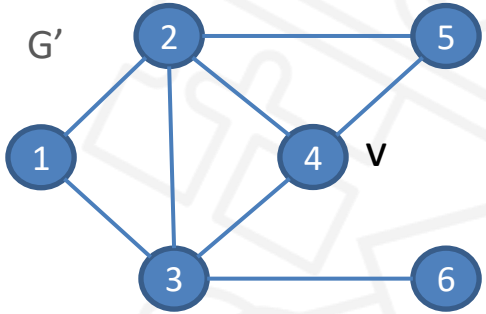




5 / 6

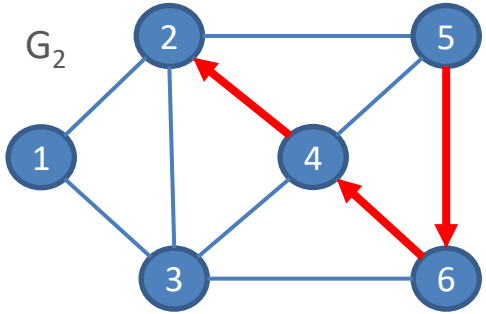
Método de Fleury – Exemplo 2

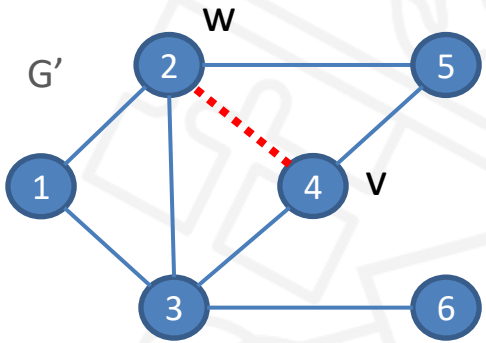




5 / 6 / 4

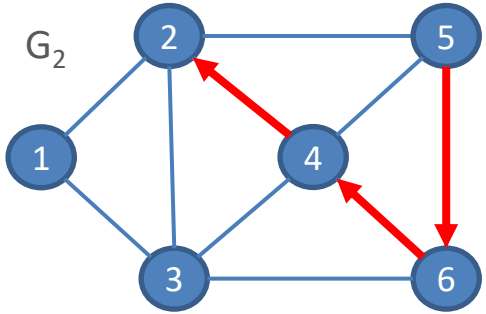
Método de Fleury – Exemplo 2

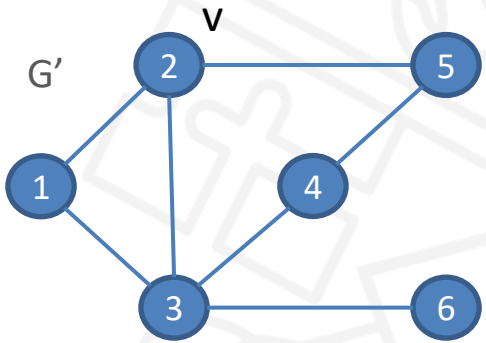




5 / 6 / 4

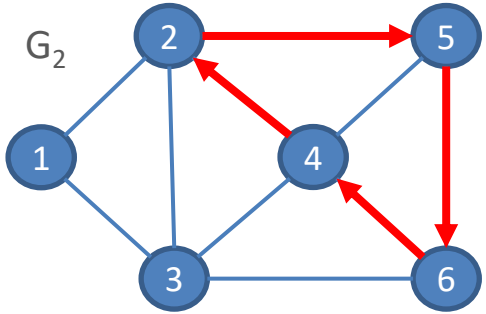
Método de Fleury – Exemplo 2

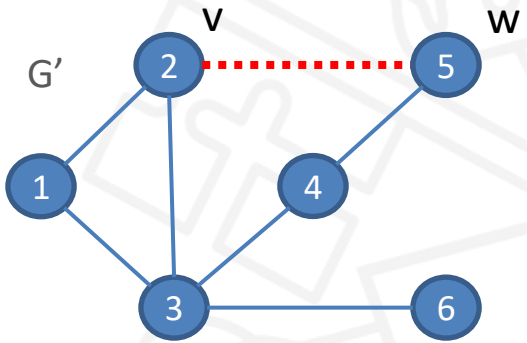




5 / 6 / 4 / 2

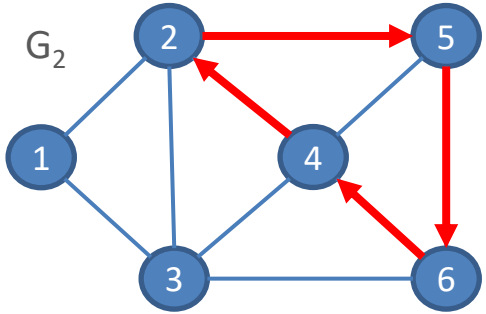
Método de Fleury – Exemplo 2

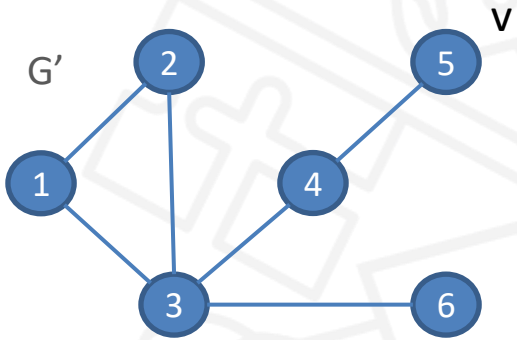




5 / 6 / 4 / 2

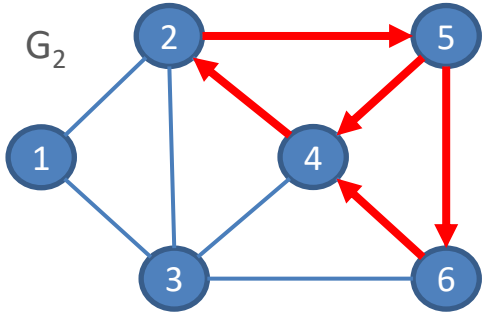
Método de Fleury – Exemplo 2

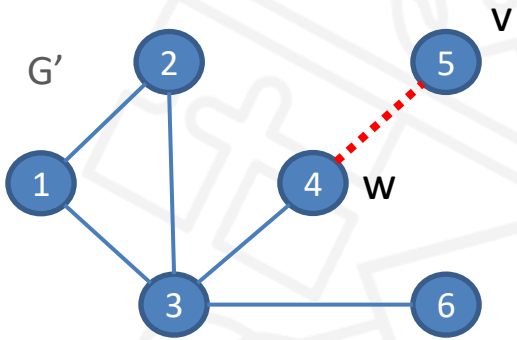




5 / 6 / 4 / 2 / 5

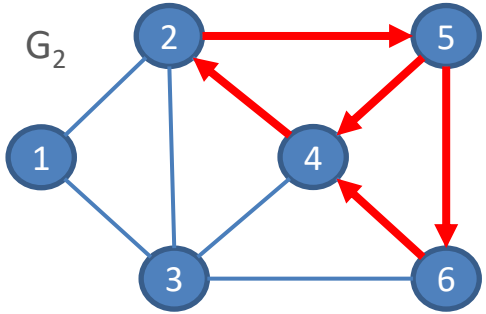
Método de Fleury – Exemplo 2

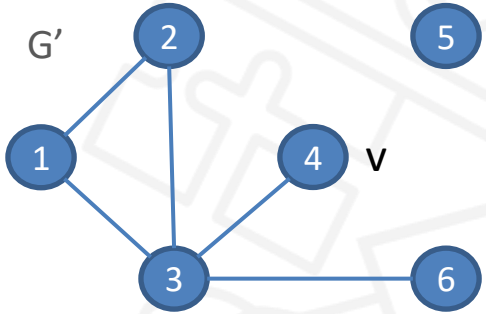




5 / 6 / 4 / 2 / 5

Método de Fleury – Exemplo 2





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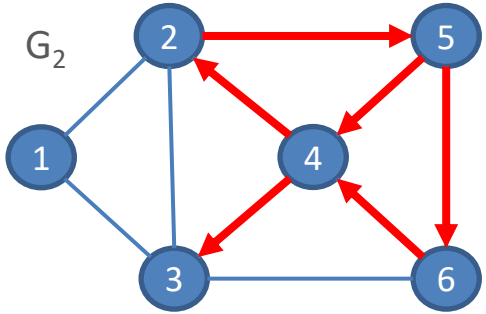
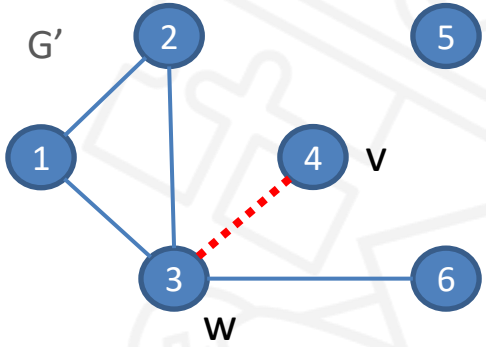
Método de Fleury – Exemplo 2





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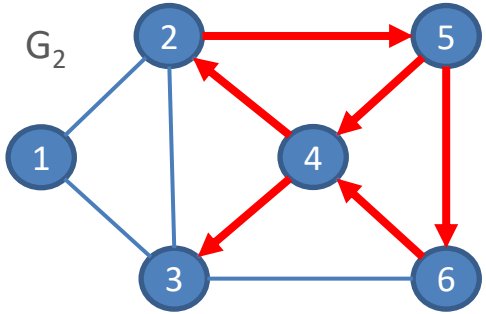
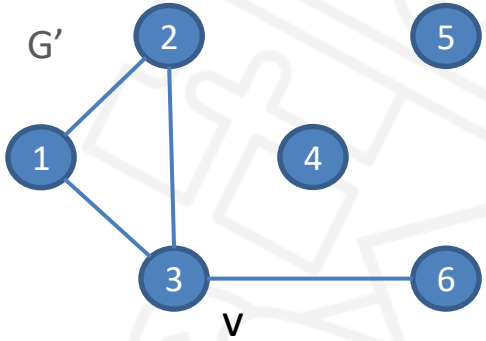
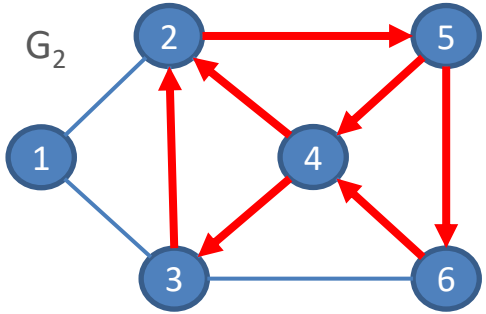
Método de Fleury – Exemplo 2

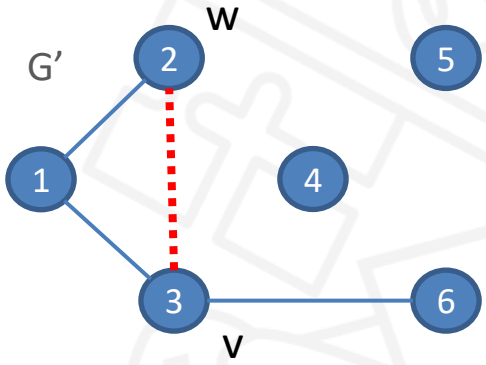




5 / 6 / 4 / 2 / 5 / 4 / 3

Método de Fleury – Exemplo 2





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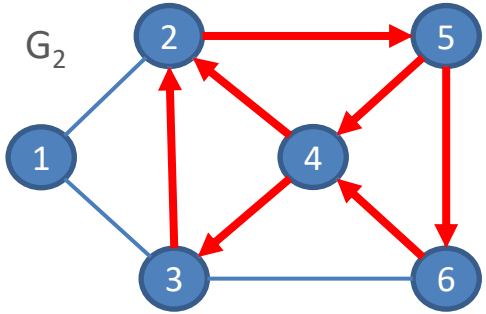
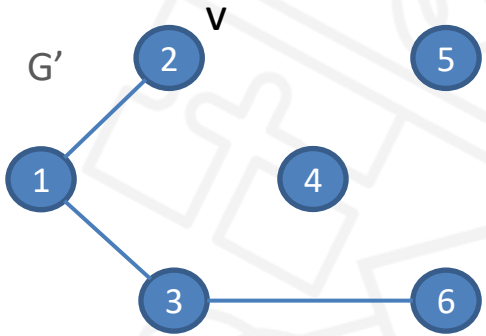
Método de Fleury – Exemplo 2





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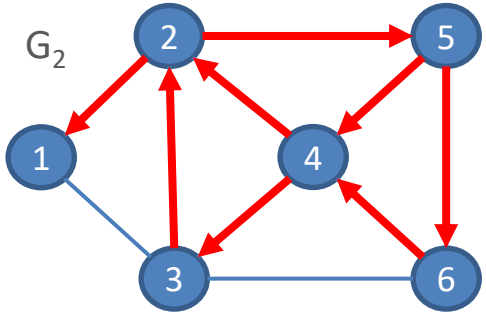
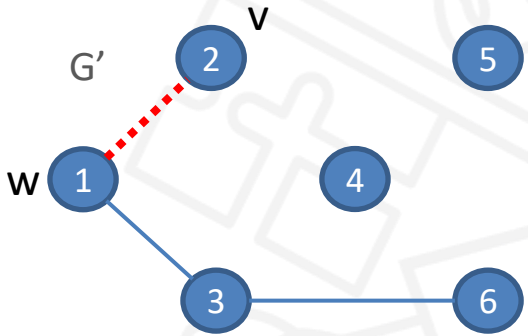
Método de Fleury – Exemplo 2





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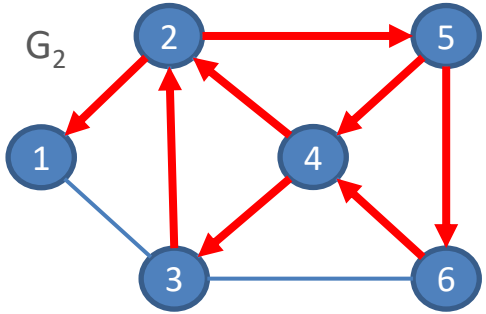
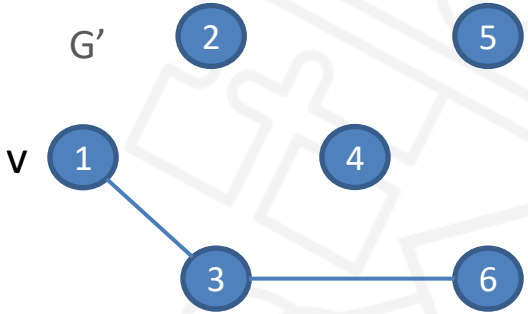
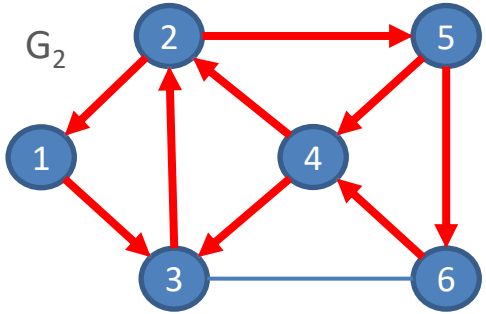
Método de Fleury – Exemplo 2

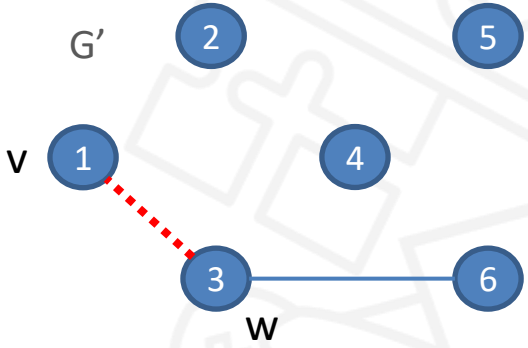




5 / 6 / 4 / 2 / 5 / 4 / 3 / 2 / 1

Método de Fleury – Exemplo 2





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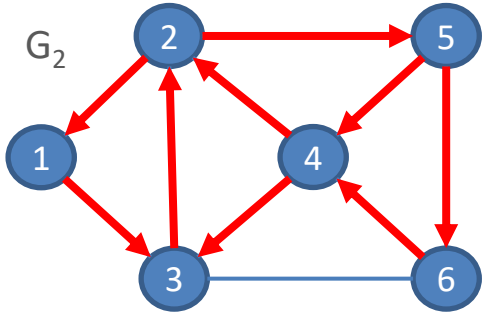
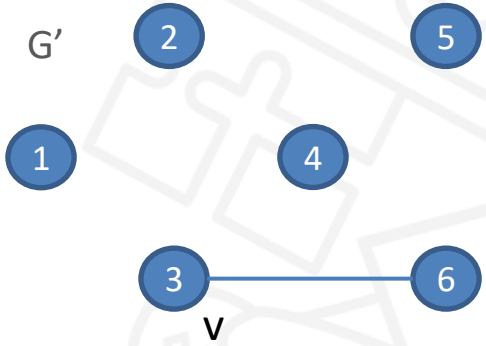
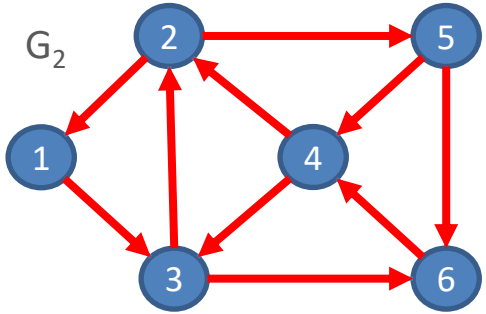
Método de Fleury – Exemplo 2

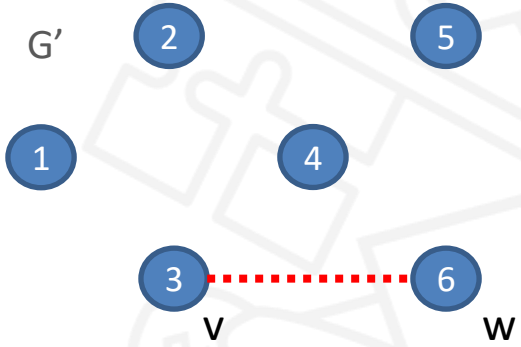




5 / 6 / 4 / 2 / 5 / 4 / 3 / 2 / 1 / 3

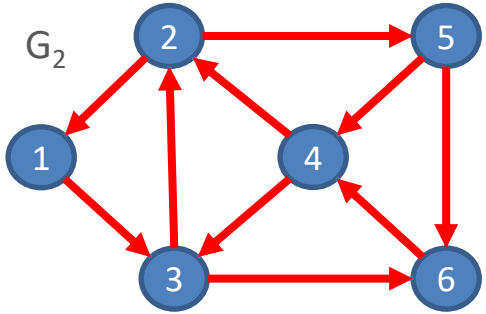
Método de Fleury – Exemplo 2





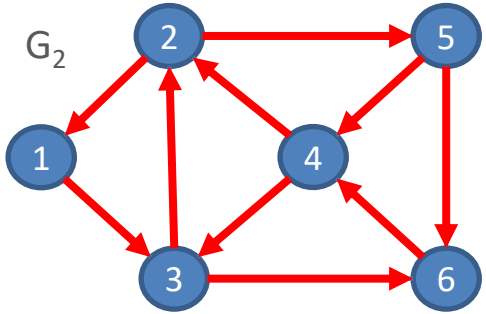
5 / 6 / 4 / 2 / 5 / 4 / 3 / 2 / 1 / 3

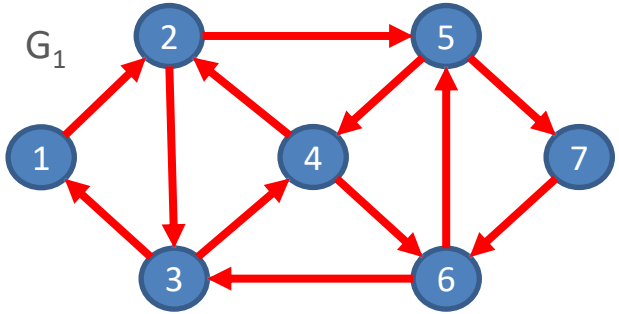
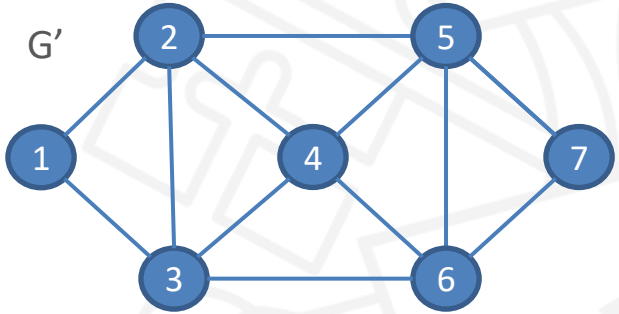
Método de Fleury – Exemplo 2



5 / 6 / 4 / 2 / 5 / 4 / 3 / 2 / 1 / 3 / 6

Método de Fleury – Exemplo 2



5 / 6 / 4 / 2 / 5 / 4 / 3 / 2 / 1 / 3 / 6 → **Trajeto euleriano**

Método de Fleury – Exemplo